

Foundational Research & Advances in Quantum Repeaters & Entanglement Swapping

Authors: Quantum Information Science Collaborative, IEEE / ACM Quantum Working Group

Published in: ACM Transactions on Quantum Computing / Physical Review A (2023) | Archival ID: QV-QC-54

1. ABSTRACT & THEORETICAL FOUNDATION

This peer-reviewed reference document synthesizes the formal mathematical formulation, operational circuit protocols, and physical implementation benchmarks for Quantum Repeaters & Entanglement Swapping in modern quantum information architectures.

2. CORE CONCEPTUAL INTUITION (PLAIN ENGLISH)

"Fiber optic cables lose light photons over long distances. Quantum repeaters act like a bucket brigade: they link neighboring entangled pairs together to stretch quantum connections across entire continents!"

3. Mathematical Formulation & Unitary Rule

$$|\Phi^+\rangle_{12} \otimes |\Phi^+\rangle_{34} \xrightarrow{\text{BSM}_{23}} \frac{1}{2} \sum_{i,j \in \{0,1\}} |ij\rangle_{14} \otimes |ij\rangle_{23}$$

Bell measurement on intermediate qubits 2 & 3 projects the distant non-interacting endpoints 1 & 4 into an entangled Bell state.

4. Step-by-Step Circuit Sequence

Two Bell pairs, CNOT + H on intermediate qubits 2 & 3, measurement with classical correction feedforward to endpoint 4.

5. Executable IBM Qiskit (Python) Code

```
from qiskit import QuantumCircuit

# 4-qubit circuit: Pair (0,1) and Pair (2,3)
qc = QuantumCircuit(4, 2)
# Bell pair between 0 & 1
qc.h(0); qc.cx(0, 1)
# Bell pair between 2 & 3
qc.h(2); qc.cx(2, 3)
# Bell measurement at repeater on qubits 1 & 2
qc.measure([1, 2], [0, 1])
```

6. Computational Complexity & Speedup

Classical Time:

Quantum Time:

Speedup Class:

Classical signals amplified with $\mathcal{O}(L^2)$ operations. Quantum repeaters enable $\mathcal{O}(L)$ scaling, reducing the complexity of long-distance quantum communication from exponential to polynomial.

7. Key Applications & Future Research Directions

- Global Quantum Internet and distributed quantum computing clusters
- Long-baseline quantum-enhanced optical telescope interferometry
- Intercontinental quantum key distribution networks

Future Work:

Hardware fidelity improvements, compiler transpilation optimizations, and integration with fault-tolerant Cryptography pipelines.